

BILKENT UNIVERSITY
ELECTRICAL AND ELECTRONICS ENGINEERING

MATH 241
ENGINEERING MATHEMATICS I

MIDTERM EXAM I

3-11-2012

Instructions:

- There are five questions in the exam. They have equal weights. Solve all.
- Duration of the exam is 150 minutes.
- The exam is closed book, closed notes.
- No calculators are allowed.
- Please read each question carefully and explain your answers clearly in order to get credit.
- Do not detach pages.

Surname:.....

Name:.....

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Section:.....

Signature:.....

Question	Grade
1	
2	
3	
4	
5	
Total:	

Q1. (20 points) Find a basis for the solution space of the following linear system of equations:

$$x_1 + 3x_2 - 2x_3 + 6x_4 + 5x_5 = 0$$

$$2x_1 + 5x_2 - x_3 + 11x_4 + 8x_5 = 0$$

$$x_1 + 7x_2 - 14x_3 + 11x_4 + x_5 = 0$$

$$\left[\begin{array}{ccccc|c} 1 & 3 & -2 & 6 & 5 & 0 \\ 2 & 5 & -1 & 11 & 8 & 0 \\ 1 & 7 & -14 & 11 & 1 & 0 \end{array} \right] \xrightarrow{\substack{-2R_1 + R_2 \rightarrow R_2 \\ -R_1 + R_3 \rightarrow R_3}} \left[\begin{array}{ccccc|c} 1 & 3 & -2 & 6 & 5 & 0 \\ 0 & -1 & 3 & -1 & -2 & 0 \\ 0 & 4 & -12 & 5 & -4 & 0 \end{array} \right]$$

$$\xrightarrow{4R_2 + R_3 \rightarrow R_3} \left[\begin{array}{ccccc|c} \textcircled{1} & 3 & -2 & 6 & 5 & 0 \\ 0 & \textcircled{-1} & 3 & -1 & -2 & 0 \\ 0 & 0 & 0 & \textcircled{1} & -12 & 0 \end{array} \right] \xrightarrow{5 \text{ points}} \left[\begin{array}{ccccc|c} 1 & 0 & 7 & 0 & 35 & 0 \\ 0 & 1 & -3 & 0 & 14 & 0 \\ 0 & 0 & 0 & 1 & -12 & 0 \end{array} \right]$$

$$\text{Let } \boxed{x_5 = t} \text{ \& } \boxed{x_3 = s} \Rightarrow \boxed{x_4 = 12t}$$

$$x_2 = 3x_3 - x_4 - 2x_5 \Rightarrow x_2 = 3s - 12t - 2t \Rightarrow \boxed{x_2 = 3s - 14t}$$

$$x_1 = -3x_2 + 2x_3 - 6x_4 - 5x_5 \Rightarrow x_1 = -3(3s - 14t) + 2s - 6(12t) - 5t$$

$$\Rightarrow x_1 = -9s + 42t + 2s - 72t - 5t$$

$$\Rightarrow \boxed{x_1 = -35t - 7s}$$

2 points for each solution
(x_1, x_2, x_3, x_4, x_5)

$$\underline{x} = s \begin{bmatrix} -7 \\ 3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -35 \\ -14 \\ 0 \\ 12 \\ 1 \end{bmatrix}$$

$$\Rightarrow \text{A basis is } \left\{ \begin{bmatrix} -7 \\ 3 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -35 \\ -14 \\ 0 \\ 12 \\ 1 \end{bmatrix} \right\} \quad 5 \text{ points}$$

Q2. (20 points) In $\mathbb{R}^2$, vector addition and scalar multiplication are defined as follows:

$$\forall \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \in \mathbb{R}^2 \text{ and } c \in \mathbb{R}, \mathbf{x} + \mathbf{y} = \begin{bmatrix} x_1 + y_1 + 1 \\ x_2 + y_2 + 1 \end{bmatrix} \text{ and } c\mathbf{x} = \begin{bmatrix} cx_1 + c - 1 \\ cx_2 + c - 1 \end{bmatrix}$$

Prove or disprove that $\mathbb{R}^2$ together with the above defined operations is a vector space.

* Closureness will be satisfied because the new vectors from $\underline{x} + \underline{y}$ & $c\underline{x}$ will also be in $\mathbb{R}^2$.

(a) Commutativity: $\underline{x} + \underline{y} = \begin{bmatrix} x_1 + y_1 + 1 \\ x_2 + y_2 + 1 \end{bmatrix} = \begin{bmatrix} y_1 + x_1 + 1 \\ y_2 + x_2 + 1 \end{bmatrix} = \underline{y} + \underline{x}$ ✓

(b) Associativity: $(\underline{x} + \underline{y}) + \underline{z} = \begin{bmatrix} x_1 + y_1 + 1 \\ x_2 + y_2 + 1 \end{bmatrix} + \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} x_1 + y_1 + z_1 + 2 \\ x_2 + y_2 + z_2 + 2 \end{bmatrix}$

$$= \begin{bmatrix} x_1 + y_1 + z_1 + 1 + 1 \\ x_2 + y_2 + z_2 + 1 + 1 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} y_1 + z_1 + 1 \\ y_2 + z_2 + 1 \end{bmatrix} = \underline{x} + (\underline{y} + \underline{z})$$
 ✓

(c) $\underline{x} + \underline{0} = \underline{x} \Rightarrow \text{let } \underline{0} = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \Rightarrow \begin{bmatrix} x_1 + \alpha + 1 \\ x_2 + \beta + 1 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \Rightarrow \begin{matrix} \alpha = -1 \\ \beta = -1 \end{matrix}$

$$\Rightarrow \underline{0} = \begin{bmatrix} -1 \\ -1 \end{bmatrix} //$$

(d) let $-\underline{x} = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \Rightarrow \underline{x} + (-\underline{x}) = \underline{0} \Rightarrow \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} x_1 + \alpha + 1 \\ x_2 + \beta + 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$

$$\Rightarrow \begin{matrix} \alpha = -x_1 - 2 \\ \beta = -x_2 - 2 \end{matrix} \Rightarrow -\underline{x} = \begin{bmatrix} -x_1 - 2 \\ -x_2 - 2 \end{bmatrix} //$$

(e) $\alpha(\beta \underline{x}) = \alpha \begin{bmatrix} \beta x_1 + \beta - 1 \\ \beta x_2 + \beta - 1 \end{bmatrix} = \begin{bmatrix} \alpha\beta x_1 + \alpha\beta - \alpha + \alpha - 1 \\ \alpha\beta x_2 + \alpha\beta - \alpha + \alpha - 1 \end{bmatrix} = \begin{bmatrix} \alpha\beta x_1 + \alpha\beta - 1 \\ \alpha\beta x_2 + \alpha\beta - 1 \end{bmatrix} = (\alpha\beta) \underline{x}$ ✓

(f) $(\alpha + \beta) \underline{x} = \begin{bmatrix} (\alpha + \beta) x_1 + \alpha + \beta - 1 \\ (\alpha + \beta) x_2 + \alpha + \beta - 1 \end{bmatrix}$ — (A)

$$\alpha \underline{x} = \begin{bmatrix} \alpha x_1 + \alpha - 1 \\ \alpha x_2 + \alpha - 1 \end{bmatrix}, \beta \underline{x} = \begin{bmatrix} \beta x_1 + \beta - 1 \\ \beta x_2 + \beta - 1 \end{bmatrix} \Rightarrow \alpha \underline{x} + \beta \underline{x} = \begin{bmatrix} \alpha x_1 + \alpha - 1 + \beta x_1 + \beta - 1 + 1 \\ \alpha x_2 + \alpha - 1 + \beta x_2 + \beta - 1 + 1 \end{bmatrix}$$

$$\Rightarrow \alpha \underline{x} + \beta \underline{x} = \begin{bmatrix} (\alpha + \beta) x_1 + \alpha + \beta - 1 \\ (\alpha + \beta) x_2 + \alpha + \beta - 1 \end{bmatrix} \text{ — (B)} \quad (A) = (B) \quad \checkmark$$

$$(g) \alpha(\underline{x} + \underline{y}) = \alpha \begin{bmatrix} x_1 + y_1 + 1 \\ x_2 + y_2 + 1 \end{bmatrix} = \begin{bmatrix} \alpha x_1 + \alpha y_1 + \alpha + \alpha - 1 \\ \alpha x_2 + \alpha y_2 + \alpha + \alpha - 1 \end{bmatrix} = \begin{bmatrix} \alpha(x_1 + y_1) + 2\alpha - 1 \\ \alpha(x_2 + y_2) + 2\alpha - 1 \end{bmatrix} \quad (4)$$

$$\alpha \underline{x} = \begin{bmatrix} \alpha x_1 + \alpha - 1 \\ \alpha x_2 + \alpha - 1 \end{bmatrix}, \quad \alpha \underline{y} = \begin{bmatrix} \alpha y_1 + \alpha - 1 \\ \alpha y_2 + \alpha - 1 \end{bmatrix} \Rightarrow \alpha \underline{x} + \alpha \underline{y} = \begin{bmatrix} \alpha(x_1 + y_1) + 2\alpha - 1 \\ \alpha(x_2 + y_2) + 2\alpha - 1 \end{bmatrix} \quad (5)$$

$$(A) = (B) \quad \checkmark$$

$$(h) 1\underline{x} = 1 \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 + 1 - 1 \\ x_2 + 1 - 1 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \underline{x} \quad \checkmark$$

One can also check the following:

$$-\underline{0} = \underline{0} \quad \text{because } \underline{0} = \begin{bmatrix} -1 \\ -1 \end{bmatrix} \Rightarrow -\underline{0} = \begin{bmatrix} -(-1) - 2 \\ -(-1) - 2 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix} \quad \checkmark$$

$$0\underline{u} = \underline{0} \quad \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0u_1 + 0 - 1 \\ 0u_2 + 0 - 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix} = \underline{0} \quad \checkmark$$

$$\alpha \underline{0} = \alpha \begin{bmatrix} -1 \\ -1 \end{bmatrix} = \begin{bmatrix} -\alpha + \alpha - 1 \\ -\alpha + \alpha - 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix} = \underline{0} \quad \checkmark$$

$$(-1)\underline{u} = (-1) \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} -u_1 - 1 - 1 \\ -u_2 - 1 - 1 \end{bmatrix} = \begin{bmatrix} -u_1 - 2 \\ -u_2 - 2 \end{bmatrix} = \underline{-u} \quad \checkmark$$

Q3. (20 points) Let $\mathbf{v}_1$ and $\mathbf{v}_2$ be two linearly independent (LI) vectors. It is possible to obtain two orthonormal vectors $\hat{\mathbf{e}}_1$ and $\hat{\mathbf{e}}_2$ from $\mathbf{v}_1$ and $\mathbf{v}_2$. Then, we can write $\hat{\mathbf{e}}_1 = c_1 \mathbf{v}_1$ and $\hat{\mathbf{e}}_2 = c_2 \mathbf{v}_1 + c_3 \mathbf{v}_2$. Clearly specify c_1, c_2 and c_3 in terms of $\mathbf{v}_1$ and $\mathbf{v}_2$.

$$* \hat{\mathbf{e}}_1 = \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} = c_1 \mathbf{v}_1 \Rightarrow c_1 = \frac{1}{\|\mathbf{v}_1\|} \text{ and note that } \|\mathbf{v}_1\| \neq 0 \text{ because } \mathbf{v}_1 \neq \mathbf{0}$$

$$* \hat{\mathbf{e}}_2 = \frac{\mathbf{v}_2 - (\mathbf{v}_2 \cdot \hat{\mathbf{e}}_1) \hat{\mathbf{e}}_1}{\|\mathbf{v}_2 - (\mathbf{v}_2 \cdot \hat{\mathbf{e}}_1) \hat{\mathbf{e}}_1\|} \quad \text{first note that } \|\mathbf{v}_2 - (\mathbf{v}_2 \cdot \hat{\mathbf{e}}_1) \hat{\mathbf{e}}_1\| \neq 0$$

$$\text{because if } \|\mathbf{v}_2 - (\mathbf{v}_2 \cdot \hat{\mathbf{e}}_1) \hat{\mathbf{e}}_1\| = 0 \Rightarrow \mathbf{v}_2 - (\mathbf{v}_2 \cdot \hat{\mathbf{e}}_1) \hat{\mathbf{e}}_1 = \mathbf{0} \Rightarrow \mathbf{v}_2 = (\mathbf{v}_2 \cdot \hat{\mathbf{e}}_1) \hat{\mathbf{e}}_1$$

$$\Rightarrow \mathbf{v}_2 = (\mathbf{v}_2 \cdot \hat{\mathbf{e}}_1) \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} = \underbrace{(\mathbf{v}_2 \cdot \hat{\mathbf{e}}_1)}_{\text{a scalar}} \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} \quad \text{implies that } \mathbf{v}_2 \text{ is linearly dependent to } \mathbf{v}_1.$$

$$\text{But because it is given that } \mathbf{v}_1 \text{ and } \mathbf{v}_2 \text{ are LI } \|\mathbf{v}_2 - (\mathbf{v}_2 \cdot \hat{\mathbf{e}}_1) \hat{\mathbf{e}}_1\| \neq 0$$

$$\text{let } \beta = \|\mathbf{v}_2 - (\mathbf{v}_2 \cdot \hat{\mathbf{e}}_1) \hat{\mathbf{e}}_1\|$$

$$\Rightarrow \hat{\mathbf{e}}_2 = \frac{1}{\beta} \mathbf{v}_2 - \frac{1}{\beta} (\mathbf{v}_2 \cdot \hat{\mathbf{e}}_1) \hat{\mathbf{e}}_1 \quad \text{but } \hat{\mathbf{e}}_1 = \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|}$$

$$\Rightarrow \hat{\mathbf{e}}_2 = \frac{1}{\left\| \mathbf{v}_2 - \frac{(\mathbf{v}_2 \cdot \mathbf{v}_1) \mathbf{v}_1}{\|\mathbf{v}_1\|^2} \right\|} \mathbf{v}_2 - \frac{1}{\left\| \mathbf{v}_2 - \frac{(\mathbf{v}_2 \cdot \mathbf{v}_1) \mathbf{v}_1}{\|\mathbf{v}_1\|^2} \right\|} \frac{(\mathbf{v}_2 \cdot \mathbf{v}_1) \mathbf{v}_1}{\|\mathbf{v}_1\|^2}$$

$$\Rightarrow \boxed{c_2 = \frac{1}{\left\| \mathbf{v}_2 - \frac{(\mathbf{v}_2 \cdot \mathbf{v}_1) \mathbf{v}_1}{\|\mathbf{v}_1\|^2} \right\|}} \quad \boxed{c_3 = -\frac{\mathbf{v}_2 \cdot \mathbf{v}_1}{\left\| \mathbf{v}_2 - \frac{(\mathbf{v}_2 \cdot \mathbf{v}_1) \mathbf{v}_1}{\|\mathbf{v}_1\|^2} \right\| \|\mathbf{v}_1\|^2}}$$

$$\boxed{c_3 = \frac{1}{\left\| \mathbf{v}_2 - \frac{(\mathbf{v}_2 \cdot \mathbf{v}_1) \mathbf{v}_1}{\|\mathbf{v}_1\|^2} \right\|}}$$

Q4. Let S be the set of all $n \times n$ diagonal matrices with real entries and zero trace, i.e.,

$$S = \left\{ \mathbf{A} \in S : \mathbf{A} = \{a_{ij}\}_{n \times n} \text{ such that } a_{ij} = 0 \text{ if } i \neq j \text{ and } \text{trace}(\mathbf{A}) = 0 \right\}.$$

Note that $\text{trace}(\mathbf{A}) \triangleq \sum_{i=1}^n a_{ii}$.

- (a) (10 points) Show that S is a subspace of the vector space spanned by all $n \times n$ matrices with real entries, (assuming that standard matrix addition and scalar multiplication are used).
- (b) (10 points) What is the dimension of S ?

(a) It is enough to show closure. Let $\underline{\mathbf{A}} \in S$ & $\underline{\mathbf{B}} \in S$

$$\underline{\mathbf{A}} + \underline{\mathbf{B}} \text{ is also a diagonal matrix and } \text{trace}(\underline{\mathbf{A}} + \underline{\mathbf{B}}) = \text{trace}(\underline{\mathbf{A}}) + \text{trace}(\underline{\mathbf{B}}) = 0 \\ \Rightarrow \underline{\mathbf{A}} + \underline{\mathbf{B}} \in S$$

$$\alpha \in \mathbb{R} \Rightarrow \alpha \underline{\mathbf{A}} \text{ is also a diagonal matrix and } \text{trace}(\alpha \underline{\mathbf{A}}) = \alpha \text{trace}(\underline{\mathbf{A}}) = 0 \\ \Rightarrow \alpha \underline{\mathbf{A}} \in S$$

$\therefore S$ is a subspace

(b) $\underline{\mathbf{A}} = \begin{bmatrix} a_{11} & & 0 \\ & a_{22} & \\ 0 & & \ddots \\ & & & a_{n-1,n-1} & \\ & & & & a_{nn} \end{bmatrix}$ such that $a_{11} + a_{22} + \dots + a_{nn} = 0$

$$\Rightarrow a_{nn} = -a_{11} - a_{22} - \dots - a_{n-1,n-1}$$

$\underbrace{\hspace{10em}}_{n-1 \text{ free variables}}$

$$\Rightarrow \underline{\mathbf{A}} = \begin{bmatrix} a_{11} & & 0 \\ & a_{22} & \\ 0 & & \ddots \\ & & & -a_{11} - a_{22} - \dots - a_{n-1,n-1} \end{bmatrix} = a_{11} \begin{bmatrix} 1 & & 0 \\ & 0 & \\ 0 & & \ddots \\ & & & 0 & \\ & & & & -1 \end{bmatrix} + a_{22} \begin{bmatrix} 0 & 1 & 0 \\ & 0 & \\ 0 & & \ddots \\ & & & 0 & \\ & & & & -1 \end{bmatrix} + \dots$$

$$+ a_{n-1,n-1} \begin{bmatrix} 0 & & 0 \\ & \ddots & \\ 0 & & 0 & 1 & \\ & & & & -1 \end{bmatrix}$$

$$\Rightarrow \dim(S) = n-1$$

Q5. For each of the following statements, provide a complete proof if the statement is true, otherwise provide a counter example.

(a)* (5 points) Let S be the set of all vectors $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ in $\mathbb{R}^2$ such that $x_1 = 1$. Then, the span of S is $\mathbb{R}^2$.

(b) (5 points) If S_1 and S_2 are two subspaces of the vector space V , then $S_1 \cap S_2$ is also a subspace of V .

(c) (5 points) In $\mathbb{R}^2$ for $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ and $\mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$, the following mapping is given:

$$\langle \mathbf{v}, \mathbf{w} \rangle = v_1 w_1 + v_1 w_2 + v_2 w_1 + v_2 w_2.$$

This mapping is a valid inner (i.e., dot) product in $\mathbb{R}^2$.

(d) (5 points) Let $\mathbf{u}$ and $\mathbf{v}$ be arbitrary non-zero vectors such that $\mathbf{u} \in \mathbb{R}^n$ and $\mathbf{v} \in \mathbb{R}^m$ where both n and m are greater than 2. Let $\mathbf{A} = \mathbf{u}\mathbf{v}^T$. Then $\mathbf{A}\mathbf{x} = \mathbf{0}$ has infinitely many solutions.

(a) TRUE let $\mathbf{v}_1 = \begin{bmatrix} 1 \\ a \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} 1 \\ b \end{bmatrix}$, then $\mathbf{v}_1$ and $\mathbf{v}_2$ are L.I. as long as $a \neq b$.

Because $\dim(\mathbb{R}^2) = 2$ a basis in $\mathbb{R}^2$ should have 2 L.I. vectors.

OR let $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \in \mathbb{R}^2$ and consider $\begin{bmatrix} 1 \\ a \end{bmatrix}$ and $\begin{bmatrix} 1 \\ b \end{bmatrix}$ where $a \neq b$ & $a, b \in \mathbb{R}$

$$\Rightarrow \mathbf{y} = \alpha \begin{bmatrix} 1 \\ a \end{bmatrix} + \beta \begin{bmatrix} 1 \\ b \end{bmatrix} \Rightarrow \begin{cases} y_1 = \alpha + \beta \\ y_2 = \alpha a + \beta b \end{cases} \Rightarrow \beta = \frac{y_2 - \alpha a}{b - a}, \alpha = y_1 - \frac{y_2 - \alpha a}{b - a}$$

so α and β are uniquely determined provided that $a \neq b$.

(b) TRUE We have to show that if $\mathbf{u}_1 \in S_1 \cap S_2$ and $\mathbf{u}_2 \in S_1 \cap S_2$

then $\forall \alpha_1, \alpha_2 \in \mathbb{R} \quad \alpha_1 \mathbf{u}_1 + \alpha_2 \mathbf{u}_2 \in S_1 \cap S_2$

Since S_1 is a subspace, for $\mathbf{u}_1 \in S_1$ and $\mathbf{u}_2 \in S_1$ $\alpha_1 \mathbf{u}_1 + \alpha_2 \mathbf{u}_2 \in S_1$
 Since S_2 is a subspace, for $\mathbf{u}_1 \in S_2$ and $\mathbf{u}_2 \in S_2$ $\alpha_1 \mathbf{u}_1 + \alpha_2 \mathbf{u}_2 \in S_2$

$\Rightarrow \alpha_1 \mathbf{u}_1 + \alpha_2 \mathbf{u}_2 \in S_1 \cap S_2 \Rightarrow S_1 \cap S_2$ is a subspace.

(c) FALSE let $\mathbf{v} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \in \mathbb{R}^2$ and $\mathbf{v} \neq \mathbf{0}$

$$\text{but } \langle \mathbf{v}, \mathbf{v} \rangle = 1 \cdot 1 + 1 \cdot (-1) + (-1) \cdot 1 + (-1) \cdot (-1) = 0$$

(J) TRUE let $\underline{u} = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} \in \mathbb{R}^n$ and $\underline{v} = \begin{bmatrix} v_1 \\ \vdots \\ v_m \end{bmatrix} \in \mathbb{R}^m$ & $\underline{u} \neq \underline{0}$
 $\underline{v} \neq \underline{0}$

$$\underline{u} \underline{v}^T = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} [v_1 \dots v_m] = \underbrace{\begin{bmatrix} u_1 v_1 & u_1 v_2 & \dots & u_1 v_m \\ u_2 v_1 & u_2 v_2 & \dots & u_2 v_m \\ \vdots & \vdots & \ddots & \vdots \\ u_n v_1 & u_n v_2 & \dots & u_n v_m \end{bmatrix}}_{\underline{A}}_{n \times m} = \begin{bmatrix} u_1 \underline{v}^T \\ u_2 \underline{v}^T \\ \vdots \\ u_n \underline{v}^T \end{bmatrix}$$

clearly all rows of $\underline{A}$ are linearly dependent

$$\Rightarrow \underline{A} \underline{x} = \underline{0} \Rightarrow \left[\begin{array}{c|c} u_1 \underline{v}^T & 0 \\ \vdots & \vdots \\ u_n \underline{v}^T & 0 \end{array} \right] \xrightarrow[\begin{array}{l} -\frac{u_2}{u_1} R_1 + R_2 \rightarrow R_2 \\ -\frac{u_3}{u_1} R_1 + R_3 \rightarrow R_3 \\ \vdots \\ -\frac{u_n}{u_1} R_1 + R_n \rightarrow R_n \end{array}]{\begin{array}{c} -\frac{u_2}{u_1} R_1 + R_2 \rightarrow R_2 \\ -\frac{u_3}{u_1} R_1 + R_3 \rightarrow R_3 \\ \vdots \\ -\frac{u_n}{u_1} R_1 + R_n \rightarrow R_n \end{array}} \left[\begin{array}{c|c} u_1 \underline{v}^T & 0 \\ \underline{0}^T & 0 \\ \vdots & \vdots \\ \underline{0}^T & 0 \end{array} \right] \quad \text{only many solutions.}$$